expert. The reason is that by 1985 physicists realized that supersymmetry by then a central element in the structure of string theory, could actually be incorporated into string theory in not one but five different ways. Each method results in a pairing of bosonic and fermionic vibrational patterns, but the details of this pairing as well as numerous other properties of the resulting theories differ substantially. Although their names are not all that important, it's worth recording that these five supersymmetric string theories are called the Type I theory, the Type IIA theory, the Type IIB theory, the Heterotic type O(32) theory (pronounced "oh-thirty-two"), and the Heterotic type E8 × E8 theory (pronounced "e-eight times e-eight"). All the features of string theory that we have discussed to this point are valid for each of these theories—they differ only in the finer details.

Having five different versions of what is supposedly the T.O.E.—possibly the ultimate unified theory—was quite an embarrassment for string theorists. Just as there is only one true explanation for whatever happened to Amelia Earhart (regardless of whether we will ever find it), we expect the same to be true regarding the deepest, most fundamental understanding of how the world works. We live in one universe; we expect one explanation.

One suggestion for resolving this problem might be that although there are five different superstring theories, four might be ruled out simply by experiment, leaving one true and relevant explanatory framework. But even if this were the case, we would still be left with the nagging question of why the other theories exist in the first place. In the wry words of Witten, "If one of the five theories describes our universe, who lives in the other four worlds?"7 A physicist's dream is that the search for the ultimate answers will lead to a single, unique, absolutely inevitable conclusion. Ideally, the final theory—whether string theory or something else—should be the way it is because there simply is no other possibility. If we were to discover that there is only one logically sound theory incorporating the basic ingredients of relativity and quantum mechanics, many feel that we would have reached the deepest understanding of why the universe has the properties it does. In short, this would be unified-theory paradise.8

As we will see in Chapter 12, recent research has taken superstring theory one giant step closer to this unified utopia by showing that the